TOPIC 1 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

Introduction to ODEs: Classification and First Models

What a differential equation is, how to classify it (ODE or PDE, order, linear or nonlinear, single equation or system), and the first two models of the course: population growth and decay, and free fall. Sections 1 to 6 follow page 2 of the class notes item by item; Robinson formalises and extends each step, and material beyond the notes is labelled as a complement.

Ground truth: Robinson (2004) Ch. 3 (pp. 11–16), Sec. 5.3 (pp. 29–31), Sec. 7.5.2 (p. 52), Ch. 8 Sec. 8.2–8.5 (pp. 61–68) • Class notes Notes.pdf, p. 2 • Lecture note *Definition of linear ODE* (Sep 9) • Lecture map ODE_lecture_theory_to_Robinson_chapters.txt , Part I (Sep 8–10)

≈ 35 min study time • 40 displayed equations • 2 figures

SECTION 01 • NOTES P. 2 • ROBINSON SEC. 3.1 • LECTURE SEP 8

1. What a differential equation is

Physical laws are rarely statements about quantities themselves; they are statements about how quantities *change*. Newton's second law relates force to the rate of change of momentum, a mass balance relates the net inflow to the rate of change of the stored mass, and a population model relates the number of births per year to the current population. Writing any of these laws down produces an equation in which the unknown is a **function**, and that function appears together with some of its derivatives. Such an equation is a **differential equation**.

DEFINITION — DIFFERENTIAL EQUATION [NOTES, P. 2]

A **differential equation** is an equation involving an unknown function (of one or more independent variables) and its derivatives.

DEFINITION — INDEPENDENT AND DEPENDENT VARIABLES

The **independent variables** are the arguments of the unknown function (e.g. t in $x(t)$, x in $y(x)$, both x and y in $G(x, y)$). The unknown function itself is the **dependent variable**. A differential equation relates a dependent variable, its derivatives and the independent variables. [Robinson, Sec. 3.1, p. 11]

Notation. The class notes write every derivative with a prime, also in time: $x'(t) = dx/dt$. Robinson keeps the prime for derivatives with respect to x and uses a dot for derivatives with respect to time:

$$\dot{z} = \frac{dz}{dt}, \qquad \ddot{z} = \frac{d^2 z}{dt^2}, \qquad y' = \frac{dy}{dx}, \qquad y'' = \frac{d^2 y}{dx^2}.$$

Both notations appear on this page; they mean the same thing.

1.1 Ordinary versus partial differential equations

The most important distinction depends on how many independent variables there are. With one independent variable only ordinary derivatives can occur; with two or more, the derivatives are partial.

ODE AND PDE, WITH THE EXAMPLES OF THE CLASS NOTES [NOTES, P. 2]

- An **ODE** is a differential equation in which the unknown function depends on **only one** independent variable. Example: $x'(t) = x(t)$, with dependent variable x and independent variable t. The function $x(t) = e^t$ is a solution, since $\frac{d}{dt}e^t = e^t$.
- A **PDE** is a differential equation in which the unknown function depends on **more than one** variable. Example: the heat equation for $u(x, t)$,

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}.$$

(The notes write the derivatives with d; with two independent variables they are partial derivatives.)

Robinson gives the same definitions with a wider set of examples [Robinson, Sec. 3.1, pp. 11–12]:

EQUATION	NAME / MEANING	TYPE	SOURCE
$\frac{dy}{dx} = f(x)$	Prescribed slope of the graph of y	ODE, independent variable x	p. 11
$m\,\ddot{\mathbf{x}} = \mathbf{f}(t)$	Newton's second law for a particle at $\mathbf{x}(t) = (x, y, z)$	ODE (system of 3), variable t	p. 12
$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$	Schrödinger equation, complex $\psi(x)$	ODE, variable x	p. 12
$\frac{\partial h}{\partial t} = k\,\frac{\partial^2 h}{\partial x^2}$	Heat equation for the temperature $h(x, t)$ of a rod	PDE, variables x, t	p. 12
$\rho\big(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\big) - \mu\Delta\mathbf{u} + \nabla p = \mathbf{f},\ \ \nabla\cdot\mathbf{u} = 0$	Navier–Stokes equations for a fluid velocity $\mathbf{u}(x_1, x_2, x_3, t)$	PDE system, 4 variables	Eqs. (3.1)–(3.3), p. 12

AEROSPACE RELEVANCE

The Navier–Stokes equations (3.3) are the model behind every computational fluid dynamics code used to design wings, intakes and nozzles. They are PDEs; this course deals with ODEs only [Robinson, p. 12]. ODEs still appear constantly in aerospace work: flight-path and attitude dynamics (t is the only independent variable), structural vibration modes, and the self-similar reductions of the flow equations, such as the boundary-layer equation of Section 2.

2. Order and autonomy

DEFINITION — ORDER [NOTES, P. 2]

The **order** of an ODE is the order of the highest derivative in the ODE.

Robinson makes this precise [Robinson, Sec. 3.2, Eq. (3.4), p. 13]: an nth-order ODE for $y(t)$ is an equation of the form

$$F\left(\frac{d^n y}{dt^n},\ \frac{d^{n-1}y}{dt^{n-1}},\ \ldots,\ \frac{dy}{dt},\ y,\ t\right) = 0, \tag{3.4}$$

in which $d^n y/dt^n$ really occurs in F. The last condition is not pedantry. If $F(\ddot{y}, \dot{y}, y, t) = y - t$, then "$F = 0$" is the algebraic relation $y = t$, which contains no derivative at all and so is not a differential equation [p. 13]. The examples of the class notes: $x' = x$ is first order, the heat equation is second order (in x), and Newton's law $m\,y'' = F$ (Section 6) is second order.

EQUATION	PHYSICAL MEANING	ORDER
$\frac{dy}{dx} = f(x)$	Slope of a graph	1
$\frac{1}{2}m\dot{x}^2 + V(x) = E$	Energy conservation: kinetic plus potential energy is constant	1
$m\frac{d^2x}{dt^2} = F$	Newton's second law	2
$\psi''' + \frac{1}{2}\psi\psi'' = 0$	Fluid boundary layers (known in fluid mechanics as the Blasius equation for flow along a flat plate)	3

[Robinson, Sec. 3.2, p. 13]

DEFINITION — AUTONOMOUS EQUATION [ROBINSON, P. 13]

If the independent variable t does not occur explicitly in the equation, as in $\dfrac{dy}{dt} = f(y)$, the equation is **autonomous**. Otherwise it is **non-autonomous**.

In the class notes autonomy appears on p. 3, where it is used for the stability of first-order equations (Topic 2). Its physical meaning is that the law does not depend on the clock. If $y(t)$ solves $\dot{y} = f(y)$, then so does the time-shifted function $z(t) = y(t - s)$ for any fixed s. To see this, apply the chain rule with the inner function $u(t) = t - s$, $du/dt = 1$:

$$\frac{dz}{dt} = \frac{dy}{du}\Big|_{u=t-s} \cdot \frac{du}{dt} = f\big(y(t-s)\big) \cdot 1 = f\big(z(t)\big).$$

So an experiment started today and the same experiment started tomorrow evolve identically. This is what makes the geometric phase-line method of Topic 2 work.

HIGHER ORDER MEANS MORE STATE, NOT MORE DIFFICULTY IN PRINCIPLE

Any second-order equation $\ddot{x} = f(\dot{x}, x, t)$ is equivalent to a first-order system for the two unknowns $x_1 = x, x_2 = \dot{x}$:

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = f(x_2, x_1, t).$$

The first equation is the definition of x_2; the second is the original equation rewritten. The same trick ($x_k = y^{(k-1)}, k = 1, \dots, n$) turns any nth-order equation into n first-order ones. This is developed in Topic 4 [Notes, p. 9; Robinson, Sec. 25.1, p. 251 for $n = 2$; Exercise 25.1, p. 252 for general n].

SECTION 03 • NOTES P. 2 • ROBINSON SEC. 3.3 • LECTURE SEP 8–10 (CORRECTED VERSION)

3. Linear and nonlinear equations

Linearity is the classification with the largest practical consequences. Robinson's summary is blunt: linear problems are relatively "easy" (an explicit solution can usually be found) and nonlinear problems are "hard" (explicit solutions exist only in very particular cases) [p. 13]. The class notes give two things: the form of a linear equation (in the margin of p. 2) and a test to decide linearity, based on superposition.

3.1 The form of a linear ODE

LINEAR ODE [NOTES, P. 2]

An ODE of order n is **linear** if it has the form

$$a_0(x)\,y^{(n)}(x) + a_1(x)\,y^{(n-1)}(x) + \cdots + a_{n-1}(x)\,y'(x) + a_n(x)\,y(x) = b(x), \qquad \text{(L)}$$

where the coefficients a_i and the right-hand side b are functions of the independent variable only. It is **homogeneous** if $b \equiv 0$ and **inhomogeneous** otherwise.

The lecture note *Definition of linear ODE* (Sep 9) writes the same form with the indices in the opposite order, $a_n(x)\,y^{(n)} + \cdots + a_1(x)\,y' + a_0(x)\,y = b(x)$, and Robinson writes it with t as the independent variable and $f(t)$ on the right, summarising it as "only multiples of y and its derivatives occur"

[Robinson, Eq. (3.5), p. 14]. The coefficients may be as nonlinear as they like *in the independent variable*; what matters is how the *unknown* appears. Collecting the left-hand side of (L) into an operator,

$$L[y] = a_0\, y^{(n)} + a_1\, y^{(n-1)} + \cdots + a_n\, y = \sum_{j=0}^{n} a_j\, y^{(n-j)},$$

a linear ODE is simply $L[y] = b$.

3.2 The superposition test of the class notes

LINEAR / NON-LINEAR ODES [NOTES, P. 2]

Consider u_1 and u_2, two solutions to an ODE. Then the ODE is **linear** if, for all $\alpha, \beta \in \mathbb{R}$, $\alpha u_1 + \beta u_2$ is also a solution to the ODE. If not, the ODE is **non-linear**.

SCOPE OF THE TEST

"Every combination $\alpha u_1 + \beta u_2$ of solutions is a solution" is exactly the **superposition principle** of *homogeneous* linear equations. An inhomogeneous linear equation such as $x' - x = 1$ has the form (L) but fails the test as stated: with $\alpha = \beta = 0$ the zero function would have to be a solution, and $0 - 0 \neq 1$. For $b \not\equiv 0$ the test is applied to the homogeneous part $L[y] = 0$. The two examples of the notes below are tested exactly as written there.

3.3 Demonstration 1: a homogeneous linear equation passes the test

PROPOSITION — SUPERPOSITION FOR $L[y] = 0$

If $L[u_1] = 0$ and $L[u_2] = 0$, then $L[\alpha u_1 + \beta u_2] = 0$ for all $\alpha, \beta \in \mathbb{R}$.

PROOF

Set $v = \alpha u_1 + \beta u_2$.

1. Differentiation is linear: for every k, $v^{(k)} = \dfrac{d^k}{dx^k}(\alpha u_1 + \beta u_2) = \alpha u_1^{(k)} + \beta u_2^{(k)}$.

2. Substitute into L and split the sum:

$$L[v] = \sum_{j=0}^{n} a_j \left(\alpha u_1^{(n-j)} + \beta u_2^{(n-j)}\right) = \alpha \sum_{j=0}^{n} a_j u_1^{(n-j)} + \beta \sum_{j=0}^{n} a_j u_2^{(n-j)} = \alpha L[u_1] + \beta L[u_2].$$

3. With $L[u_1] = L[u_2] = 0$: $L[v] = \alpha \cdot 0 + \beta \cdot 0 = 0$. $\square$

■

EXAMPLE OF THE NOTES: IS $u' = u$ LINEAR? [NOTES, P. 2]

Let u_1 and u_2 be solutions, $u_1' = u_1$ and $u_2' = u_2$, and call $u = \alpha u_1 + \beta u_2$. Then

$$u' = \alpha u_1' + \beta u_2' = \alpha u_1 + \beta u_2 = u,$$

so u is again a solution for every α, β: the ODE is **linear**. In the form (L): $n = 1, a_0 = 1, a_1 = -1, b = 0$.

The same property gives the structure of the solutions of an inhomogeneous equation: if $L[y_p] = b$ and $L[y_h] = 0$, then $L[y_h + y_p] = L[y_h] + L[y_p] = 0 + b = b$. General solution = homogeneous solution + one particular solution (Topic 3; Robinson Sec. 14.1).

3.4 Demonstration 2: $u'' = (u')^2 + 1$ fails the test

Let u_1 and u_2 be solutions, $u_1'' = (u_1')^2 + 1$ and $u_2'' = (u_2')^2 + 1$, and let $v = \alpha u_1 + \beta u_2$ [Notes, p. 2]. By linearity of differentiation,

$$v' = \alpha u_1' + \beta u_2', \qquad v'' = \alpha u_1'' + \beta u_2''.$$

For v to be a solution we would need $v'' = (v')^2 + 1$. Compute both sides.

1. Left side, replacing u_1'' and u_2'' by their equations:

$$v'' = \alpha\big((u_1')^2 + 1\big) + \beta\big((u_2')^2 + 1\big) = \alpha\,(u_1')^2 + \beta\,(u_2')^2 + \alpha + \beta.$$

2. Right side, expanding the square $(a+b)^2 = a^2 + 2ab + b^2$:

$$(v')^2 + 1 = \big(\alpha u_1' + \beta u_2'\big)^2 + 1 = \alpha^2 (u_1')^2 + 2\alpha\beta\, u_1' u_2' + \beta^2 (u_2')^2 + 1.$$

3. Compare. The left side has α where the right side has α^2, β where it has β^2, no cross term $2\alpha\beta\, u_1' u_2'$, and the constant $\alpha + \beta$ instead of 1. The equality cannot hold for all α, β: the ODE is **non-linear**.

EXTENSION – THE CULPRIT IS THE SQUARE, NOT THE "+1"

For an inhomogeneous *linear* equation, combinations with $\alpha + \beta = 1$ are still solutions ($L[\alpha u_1 + \beta u_2] = (\alpha + \beta)\, b = b$). For $u'' = (u')^2 + 1$ even these fail. Put $\beta = 1 - \alpha$ and subtract the two sides of steps 1 and 2; since $\beta(1-\beta) = (1-\alpha)\alpha$ and $2\alpha\beta = 2\alpha(1-\alpha)$,

$$v'' - \big((v')^2 + 1\big) = \alpha(1-\alpha)(u_1')^2 + \beta(1-\beta)(u_2')^2 - 2\alpha\beta\, u_1' u_2' = \alpha(1-\alpha)\big(u_1' - u_2'\big)^2,$$

which is not zero when $0 < \alpha < 1$ and $u_1' \neq u_2'$. Such solutions exist: $w = u'$ satisfies $w' = w^2 + 1$, solved by $w = \tan(t + c)$ because $\frac{d}{dt} \tan s = \sec^2 s = \tan^2 s + 1$, so $u = -\ln\cos(t + c) + d$. Take $u_1 = -\ln\cos t$, $u_2 = -\ln\cos(t - \pi/4)$ on $-\pi/4 < t < \pi/2$ and $\alpha = \beta = \frac{1}{2}$. At $t = 0$: $u_1' = 0$, $u_2' = -1$, $u_1'' = 1$, $u_2'' = 2$, so $v'' = \frac{3}{2}$ while $(v')^2 + 1 = \frac{1}{4} + 1 = \frac{5}{4}$. The gap $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} \cdot \big(0 - (-1)\big)^2$ is the one the formula predicts.

3.5 A practical test, and the lecture examples

Read each term of the equation and ask what it does to the unknown y. The equation is linear exactly when every term is one of:

1. a function of the independent variable alone (it goes into b);
2. a function of the independent variable multiplied by y or by one derivative $y^{(k)}$, to the first power.

Anything else makes the equation nonlinear: powers such as y^2 or $(y')^2$, products such as $y\,y'$, and nonlinear functions of the unknown such as $\sin y$, e^y or $\sqrt{y'}$.

The three equations analysed in class (corrected note dated Sep 9, presented in the lecture of Sep 10):

ODE	ORDER	VERDICT	JUSTIFICATION AGAINST THE FORM (L)
$u'(t) = u(t)$	1	**Linear**, homogeneous	Rewrite as $1 \cdot u' + (-1) \cdot u = 0$: $a_0 = 1$, $a_1 = -1$, $b = 0$. Superposition: Section 3.3.
$u''(t) = \big(u'(t)\big)^2 + 1$	2	**Nonlinear**	The derivative u' is squared. No choice of $a_i(t), b(t)$ reproduces $(u')^2$. Superposition fails: Section 3.4.
$y'(x) = 4\big(y(x)\big)^2 + x^2$	1	**Nonlinear**	The unknown y itself is squared. The term x^2 is harmless: it is a function of x only and would sit in $b(x)$.

LECTURE VS. BOOK: WHY A "CORRECTED VERSION"

The lecture of Sep 8 classified these examples and the solution was flagged as needing correction; the corrected solution was given on Sep 10 together with the note *Definition of linear ODE*. The table above follows the corrected note. The third example is a useful trap: a term like x^2 looks "nonlinear", but nonlinearity in the independent variable is irrelevant; only nonlinearity in the unknown counts.

3.6 Complement (Robinson): what "solving" can mean

Robinson lists the kinds of answer one can expect, in decreasing order of explicitness [Robinson, Sec. 3.4, pp. 14-16]:

TYPE	EXAMPLE	COMMENT
Fully explicit	$y(t) = 3\cos 5t + 8\sin t$ (3.6)	Combination of elementary functions; rare.
Explicit with an integral	$y(t) = 1 + \int_0^t e^{-s^2}\,ds$ (3.7)	Still y as a function of t, but the integral has no elementary antiderivative.
Implicit	$\ln y + 4\ln x - y - 2x + 4 = 0$ (3.8)	Relates x and y without derivatives; cannot be solved for y.
Qualitative	Phase portrait of (3.9), Fig. 1.1	Describes the behaviour without any formula.

TYPE	EXAMPLE	COMMENT
Numerical	MATLAB `ode45` for $\dot{x} = t - x^2$, $x(0) = 0$	A table of approximate values at discrete times.

SECTION 04 • NOTES P. 2 • ROBINSON EQ. (3.9), CH. 25 • LECTURE SEP 8

4. Systems of ODEs

SYSTEMS OF ODES [NOTES, P. 2]

A **system of ODEs** has two or more dependent variables. Example:

$$\begin{cases} x'(t) = 2x(t) - y(t), \\ y'(t) = -x(t) + 3y(t). \end{cases}$$

The two unknowns $x(t), y(t)$ share the independent variable t. The system is **first order**, **linear** and homogeneous (each right-hand side is a combination of x and y with constant coefficients and nothing else), and autonomous. In matrix form,

$$\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

It passes the superposition test of Section 3.2: if (x_1, y_1) and (x_2, y_2) are solutions, then $(x, y) = \alpha(x_1, y_1) + \beta(x_2, y_2)$ satisfies $x' = \alpha x_1' + \beta x_2' = \alpha(2x_1 - y_1) + \beta(2x_2 - y_2) = 2x - y$, and in the same way $y' = -x + 3y$.

Preview of Topic 4: one second-order equation for x

Solve the first equation for y: $y = 2x - x'$. Differentiating, $y' = 2x' - x''$. Substitute both into the second equation:

$$2x' - x'' = -x + 3\,(2x - x') = 5x - 3x' \quad \Longrightarrow \quad x'' - 5x' + 5x = 0.$$

This is the elimination method of the class notes, p. 9, which turns $x' = ax + by$, $y' = cx + dy$ into $x'' - (a + d)x' + (ad - bc)x = 0$. Check: $a = 2, b = -1, c = -1, d = 3$ give $a + d = 5$ and $ad - bc = 6 - 1 = 5$. Second-order equations of this kind are solved in Topic 3 and systems in Topic 4.

4.1 Complement (Robinson): a nonlinear system and its phase portrait

Robinson's first example (which he interprets as two competing species only much later, p. 326) has unknowns $x(t)$ and $y(t)$ [Robinson, Eq. (3.9), p. 15]:

$$\dot{x} = x\,(4 - 2x - y), \qquad \dot{y} = y\,(9 - 3x - 3y). \tag{3.9}$$

Each species grows when rare (the constant terms $4x$ and $9y$), suffers from crowding by its own kind ($-2x^2$, $-3y^2$) and from competition with the other ($-xy$, $-3xy$). The system is first order, autonomous and nonlinear (it contains x^2, y^2 and xy), so the superposition test fails.

Where the solutions settle

A point where both right-hand sides vanish is a constant (equilibrium) solution. From $x(4 - 2x - y) = 0$ we need $x = 0$ or $y = 4 - 2x$; from $y(9 - 3x - 3y) = 0$ we need $y = 0$ or $y = 3 - x$. Combining the four options:

- $x = 0,\ y = 0$: the point $(0, 0)$;
- $x = 0,\ y = 3 - x$: the point $(0, 3)$;
- $y = 4 - 2x,\ y = 0$: the point $(2, 0)$;
- $y = 4 - 2x,\ y = 3 - x$: subtracting, $0 = 1 - x$, so $x = 1$ and $y = 2$: the point $(1, 2)$.

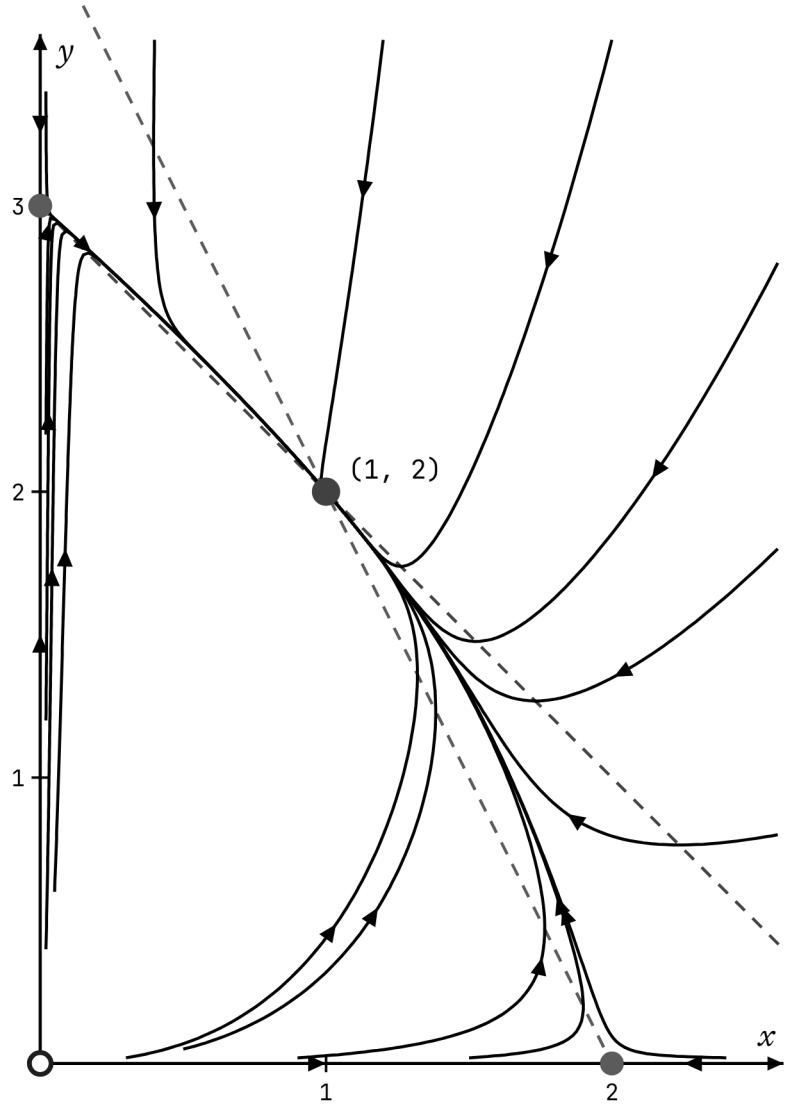


Figure 1.1. *Phase portrait of system (3.9), $\dot{x} = x(4 - 2x - y)$, $\dot{y} = y(9 - 3x - 3y)$, drawn by numerical integration (after Robinson Fig. 3.1, p. 15). Dashed lines: curves where $\dot{x} = 0$ (amber) and $\dot{y} = 0$ (cyan). Every solution starting with $x > 0$, $y > 0$ tends to the green point $(1, 2)$, although no explicit formula for $x(t)$, $y(t)$ is known.*

The figure, which Robinson calls a **phase diagram** or **phase portrait**, shows that every solution with both species present tends to $(1, 2)$ as $t \to +\infty$: the species coexist `[p. 15; species reading, p. 326]`. The vector notation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$ and the full theory of systems are the subject of Topic 4.

5. Example 1: population growth and decay

5.1 The model

POPULATION GROWTH / DECAY [NOTES, P. 2]

$x(t)$ is the population at time t, with $x(t) > 0$ for all t. The rate of change of x is proportional to x itself, with proportionality constant $k \in \mathbb{R}$. This gives the first-order linear ODE

$$x'(t) = k\,x(t), \qquad x(0) = x_0 > 0.$$

Where the law comes from. Robinson derives it from a balance [Robinson, Sec. 8.3, Eq. (8.5), p. 62]: in a short time Δt a population x gains about $\beta x\,\Delta t$ births and loses about $\delta x\,\Delta t$ deaths, with constant per-capita rates β, δ. Hence $\Delta x \approx (\beta - \delta)\,x\,\Delta t$; dividing by Δt and letting $\Delta t \to 0$ gives $x' = kx$ with $k = \beta - \delta$ (Malthus, 1798).

SYMBOL	MEANING	SI UNIT
$x(t)$	Population (number of individuals)	dimensionless count
x_0	Initial population, $x(0)$	dimensionless count
k	Net growth rate per individual	s^{-1} (in practice yr^{-1})
t	Time	s

Linearity: in the form (L) with t as variable, $1 \cdot x' + (-k) \cdot x = 0$, so $a_0 = 1, a_1 = -k, b = 0$.

5.2 Solution by integration from 0 to t

Because $x(\tau) > 0$ we may divide by it: $\dfrac{x'(\tau)}{x(\tau)} = k$. The class notes integrate both sides from 0 to t (the dummy variable is τ, so that t can be the upper limit) and evaluate the left side in two equivalent ways [Notes, p. 2].

1. **Route 1: substitution.** Put $u = x(\tau)$, so $du = x'(\tau)\,d\tau$. The limits change with it: $\tau = 0$ gives $u = x(0) = x_0$ and $\tau = t$ gives $u = x(t)$. An antiderivative of $1/u$ for $u > 0$ is $\ln u$, so by Barrow's rule

$$\int_0^t \frac{x'(\tau)}{x(\tau)}\,d\tau = \int_{x_0}^{x(t)} \frac{du}{u} = \Big[\ln u\Big]_{x_0}^{x(t)} = \ln x(t) - \ln x_0 = \ln \frac{x(t)}{x_0}.$$

The notes abbreviate this step as $\dfrac{dx}{dt} = kx \Rightarrow \displaystyle\int_0^t \frac{dx}{x} = \int_0^t k\,dt.$

2 **Route 2: recognise a derivative.** By the chain rule with outer function ln and inner function $x(\tau) > 0$,

$$\frac{d}{d\tau} \ln\big(x(\tau)\big) = \frac{1}{x(\tau)} \cdot x'(\tau),$$

so the integrand is the derivative of $\ln x(\tau)$, and Barrow's rule gives directly

$$\int_0^t \big(\ln x(\tau)\big)' \, d\tau = \ln x(t) - \ln x(0) = \ln \frac{x(t)}{x_0}.$$

3 **Right side.** An antiderivative of the constant k is $k\tau$:

$$\int_0^t k \, d\tau = \Big[k\tau\Big]_0^t = kt - k \cdot 0 = kt.$$

4 **Equate and solve.** $\ln \dfrac{x(t)}{x_0} = kt$. Exponentiate both sides: $\dfrac{x(t)}{x_0} = e^{kt}$.

SOLUTION

$$x(t) = x_0 \, e^{kt}.$$

- **Check the ODE:** $x'(t) = x_0 \, k \, e^{kt} = k \, x(t)$ (chain rule on e^{kt}).
- **Check the initial value:** $x(0) = x_0 e^0 = x_0$.
- **Dimensions:** $[kt] = \mathrm{s}^{-1} \cdot \mathrm{s} = 1$, so the exponential is dimensionless and x has the units of x_0 .

5.3 Growth or decay: the sign of k

CASE [NOTES, P. 2]	BEHAVIOUR	AS $t \to \infty$	TIME SCALE
$k > 0$	Population grows	$x(t) \to \infty$	Doubling time d: $2 = e^{kd}$, so $d = \ln 2/k$
$k < 0$	Population decays	$x(t) \to 0$	Half-life: $\frac{1}{2} = e^{kh}$, so $h = \ln 2/\lvert k\rvert$
$k = 0$	No change	$x(t) \equiv x_0$	–

Neither time scale depends on x_0: a population of a thousand and one of a million double in the same time. The decaying case is also the law of radioactive decay (Problem 1.3, plutonium-239).

COMPLEMENT (ROBINSON, SEC. 8.2): ANY STARTING TIME, ANY SIGN

The notes assume $x > 0$, natural for a population. For $\dot{x} = \lambda x$ with $x(t_0) = x_0$ of either sign, Robinson argues that $x \equiv 0$ is the only solution through $x_0 = 0$ (uniqueness by Picard's theorem, Topic 2, Section 2), so no other solution can reach 0; $x(t)$ keeps the sign of x_0, the moduli in $\ln|x|$ can be dropped, and the same computation gives $x(t) = x_0\, e^{\lambda(t-t_0)}$ [Robinson, Sec. 8.2, pp. 61-62].

COMPLEMENT: CENSUS TEST OF THE MODEL (ROBINSON SEC. 8.3, PP. 63-64)

Population of Great Britain and Ireland: 1801: 16 345 646; 1851: 27 533 755; 1901: 41 609 091 (Eq. 8.6).

Fitting k from 1801 and 1851, $x(1851) = x(1801)e^{50k}$:

$$k = \frac{\ln x(1851) - \ln x(1801)}{50} = \frac{\ln(27\,533\,755/16\,345\,646)}{50} = \frac{0.52145}{50} = 0.01043\ \text{yr}^{-1}.$$

- Prediction for 1901: $x(1801)e^{100k} = 46.4$ million (true: 41.6 million) – reasonable.
- Prediction for 2001: $x(1801)e^{200k} = 131.6$ million (true: about 63 million, UK plus Republic of Ireland) – more than twice too large.
- Doubling time: $\ln 2/0.01043 = 66.5$ years. Robinson rounds k to 0.010 first and quotes 69 years; both describe the same fit.

5.4 The logistic correction

WHY THE MODEL MUST BE CORRECTED [NOTES, P. 2]

In reality a population does not grow exponentially forever, but reaches a flat maximum: we use a **logistic correction**.

The logistic equation $x' = kx\,(1 - x/M) = kx - \frac{k}{M}x^2$ keeps the exponential rate for small populations and stops the growth at a maximum value M [Robinson, Sec. 8.5, Eq. (8.12), p. 66]. The class notes study it on p. 3, with the methods of the next block: its stationary points and their stability (Topic 2, Section 4.4) and its explicit solution by separation of variables (Topic 2, Section 6.3).

6. Example 2: Newton's second law and free fall

6.1 The model

NEWTON'S 2ND LAW, "$F = ma$": FREE FALL OF AN OBJECT UNDER g [NOTES, P. 2]

Let $y(t)$ be the height of the object, measured on a vertical axis pointing up. The only force is the weight, $F = -mg$, and the acceleration is $y''(t)$, so

$$-m\,g = m\,y''(t) \quad \Longrightarrow \quad \frac{d^2y}{dt^2} = -g,$$

a **second-order linear** ODE. The mass cancels.

Hypotheses. Constant gravity $g \approx 9.81\ \mathrm{m/s^2}$, no air resistance, motion along a vertical line: one degree of freedom, the height $y(t)$ [m]. In the form (L): $a_0 = 1, a_1 = a_2 = 0, b = -g$; the equation is inhomogeneous with constant right-hand side.

6.2 Integrating twice

The notes write the answer as $y = -\iint g\,dt\,dt$. With the antiderivatives and the limits shown:

1. Write $y'' = (y')'$. An antiderivative of the constant $-g$ is $-g\tau$, so integrating from 0 to t and using Barrow's rule,

$$y'(t) - y'(0) = \int_0^t (-g)\,d\tau = \Big[-g\tau\Big]_0^t = -gt.$$

   With $v_0 = y'(0)$: $y'(t) = v_0 - gt$.

2. An antiderivative of $v_0 - g\tau$ is $v_0\tau - \tfrac{1}{2}g\tau^2$ (check: $\frac{d}{d\tau}\big(v_0\tau - \tfrac{1}{2}g\tau^2\big) = v_0 - g\tau$), so

$$y(t) - y(0) = \int_0^t (v_0 - g\tau)\,d\tau = \Big[v_0\tau - \tfrac{1}{2}g\tau^2\Big]_0^t = v_0 t - \tfrac{1}{2}gt^2.$$

SOLUTION

$$y(t) = y_0 + v_0\,t - \tfrac{1}{2}g\,t^2, \qquad y_0 = y(0), \quad v_0 = y'(0).$$

Two integrations produce two constants, one per order of the equation. The notes take the object released from rest at the origin, $y_0 = v_0 = 0$:

$$y(t) = -\iint g \, dt \, dt = -\tfrac{1}{2} g \, t^2 .$$

- **Check:** $y' = -gt, y'' = -g$; $y(0) = 0, y'(0) = 0$.
- **Dimensions:** $[gt^2] = \mathrm{m\ s^{-2}} \cdot \mathrm{s^2} = \mathrm{m}$.
- **Mass independence:** m cancelled – without drag every body falls the same way.

6.3 Complement (Robinson): from momentum balance to $F = ma$

Newton's second law states that the change Δp in the momentum of a body equals the applied force F times the time Δt over which it acts, $\Delta p = F \, \Delta t$ [Robinson, Sec. 5.3, p. 30]. Dividing by Δt and letting $\Delta t \to 0$ (Robinson notes this step is somewhat imprecise) gives

$$\frac{dp}{dt} = F(t).$$

With $p = mv$, the product rule gives

$$\frac{dp}{dt} = \frac{d}{dt}(mv) = \frac{dm}{dt}\, v + m \, \frac{dv}{dt}.$$

For constant mass $dm/dt = 0$, so $m \, \dfrac{dv}{dt} = F(t)$: the familiar $F = ma$. (For variable-mass bodies such as rockets, $F = d(mv)/dt$ does not apply: the momentum carried away by the exhaust must be included in a balance over the whole system, which is outside this course.)

6.4 Complement (Robinson Exercise 5.6): release from a height h

Release the body from rest at height h: $y_0 = h$, $v_0 = 0$ in the solution of 6.2, so $y(t) = h - \frac{1}{2} g t^2$ and $y'(t) = -gt$. The book leaves this case as Exercise 5.6 (an apple falling from height h).

1. The impact time t^* solves $y(t^*) = 0$: $h = \frac{1}{2} g t^{*2}$, so $t^* = \sqrt{2h/g}$ (the positive root, since $t^* > 0$).
2. The impact velocity is $y'(t^*) = -g\sqrt{2h/g} = -\sqrt{2gh}$.

RESULT AND CHECKS

$$y(t) = h - \tfrac{1}{2} g t^2, \qquad y'(t) = -gt, \qquad t^* = \sqrt{\frac{2h}{g}}, \qquad |y'(t^*)| = \sqrt{2gh}.$$

- **Dimensions:** $[\sqrt{h/g}] = \sqrt{\mathrm{m}/(\mathrm{m\,s^{-2}})} = \mathrm{s}$; $[\sqrt{gh}] = \sqrt{\mathrm{m^2 s^{-2}}} = \mathrm{m/s}$.
- **Energy:** $\frac{1}{2}m\,y'(t^*)^2 = \frac{1}{2}m(2gh) = mgh$: the potential energy lost equals the kinetic energy gained .

EXAMPLE 5.4 (ROBINSON PP. 30–31; NUMBERS FROM THE CAPTION OF FIG. 5.6): A BRAKING CAR — THE SAME METHOD, HORIZONTALLY

A car of mass m moving at v_0 brakes with a constant force k. Newton: $m\dot{v} = -k$. Integrating from 0 to t, $v(t) - v_0 = \int_0^t (-k/m)\,d\tilde{t} = -kt/m$, so $v(t) = v_0 - kt/m$. It stops when $v(t_c) = 0$, i.e. $t_c = mv_0/k$. Integrating $\dot{x} = v_0 - kt/m$ from 0 to t_c:

$$x(t_c) - x(0) = \left[v_0\tilde{t} - \frac{k\tilde{t}^{\,2}}{2m}\right]_0^{t_c} = v_0 t_c - \frac{kt_c^2}{2m} = \frac{mv_0^2}{k} - \frac{mv_0^2}{2k} = \frac{mv_0^2}{2k}.$$

With Robinson's data $m = 1000$ kg, $k = 6500$ N, $v_0 = 30$ mph $= 13.41$ m/s: $t_c = 1000 \times 13.41/6500 = 2.06$ s and the stopping distance is $1000 \times 13.41^2/(2 \times 6500) = 13.8$ m (about 45 ft, as in his Fig. 5.7). Dimensions: $[mv_0^2/k] = \mathrm{kg\ m^2 s^{-2}/(kg\ m\ s^{-2})} = \mathrm{m}$.

6.5 Complement (Robinson): free fall with quadratic air drag, terminal velocity

At the speeds of a skydiver or a descending capsule, air resistance is not negligible. Robinson models it with a drag force proportional to the square of the speed and opposite to the motion [Sec. 7.5.2, Eq. (7.5), p. 52]:

$$m\frac{dv}{dt} = mg - kv|v|, \tag{7.5}$$

where now v is the *downward* velocity and $k > 0$ (units kg/m) collects air density, drag coefficient and frontal area. The term $v|v|$ (rather than v^2) makes the drag oppose the motion whatever its direction. For a body released from rest, $v \geq 0$ and the equation becomes

$$\frac{dv}{dt} = g - \frac{k}{m}v^2.$$

This equation is first order, autonomous and **nonlinear** (v^2). Its right-hand side vanishes at $v^* = \sqrt{mg/k}$, the **terminal velocity**: if the body ever reaches v^*, weight and drag balance and it stays there. Robinson's skydiver ($m = 100$ kg, $k \approx 1/3$ kg/m, $g \approx 9.8$ m/s^2) has $v^* = \sqrt{100 \times 9.8 \times 3} \approx 54.2$ m/s [p. 53].

Explicit solution (Robinson Exercise 8.6, worked here in full)

Write $w_T = \sqrt{mg/k}$, so that $k/m = g/w_T^2$ and $\dot{v} = \dfrac{g}{w_T^2}\,(w_T^2 - v^2)$. For $0 \leq v < w_T$ we may divide (separation of variables, justified in Topic 2):

$$\frac{dv}{w_T^2 - v^2} = \frac{g}{w_T^2}\,dt.$$

1 Partial fractions: since $w_T^2 - v^2 = (w_T - v)(w_T + v)$,

$$\frac{1}{w_T^2 - v^2} = \frac{1}{2w_T}\left(\frac{1}{w_T - v} + \frac{1}{w_T + v}\right).$$

Check: $\dfrac{(w_T + v) + (w_T - v)}{2w_T(w_T - v)(w_T + v)} = \dfrac{2w_T}{2w_T(w_T^2 - v^2)}$.

2 Antiderivatives. For $\int \frac{dv}{w_T - v}$ substitute $u = w_T - v$, $du = -dv$: $\int \frac{-du}{u} = -\ln|u| = -\ln(w_T - v)$ (here $w_T - v > 0$). Directly, $\int \frac{dv}{w_T + v} = \ln(w_T + v)$. Hence

$$\int \frac{dv}{w_T^2 - v^2} = \frac{1}{2w_T}\ln\frac{w_T + v}{w_T - v}.$$

3 Barrow's rule between $v(0) = 0$ and $v(t)$ on the left, 0 and t on the right:

$$\frac{1}{2w_T}\left[\ln\frac{w_T + v}{w_T - v}\right]_0^{v(t)} = \frac{1}{2w_T}\left(\ln\frac{w_T + v(t)}{w_T - v(t)} - \ln 1\right) = \frac{g}{w_T^2}\left[\tilde{t}\right]_0^t = \frac{gt}{w_T^2}.$$

4 Multiply by $2w_T$ and exponentiate: $\dfrac{w_T + v}{w_T - v} = e^{2gt/w_T}$. Solve for v: $w_T + v = (w_T - v)e^{2gt/w_T}$, so $v\left(1 + e^{2gt/w_T}\right) = w_T\left(e^{2gt/w_T} - 1\right)$ and, dividing numerator and denominator by e^{gt/w_T},

$$v(t) = w_T\,\frac{e^{gt/w_T} - e^{-gt/w_T}}{e^{gt/w_T} + e^{-gt/w_T}} = w_T\tanh\!\left(\frac{gt}{w_T}\right).$$

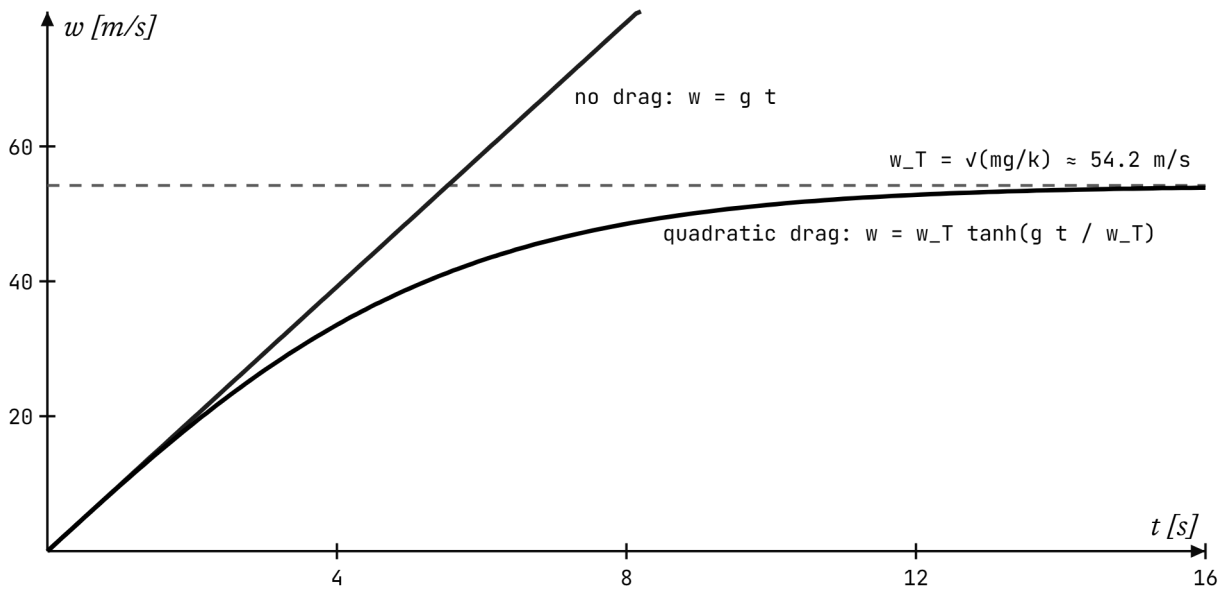


Figure 1.2. *Downward speed of a skydiver released from rest, with Robinson's data $m = 100$ kg, $k \approx 1/3$ kg/m, $g \approx 9.8$ m/s^2 (Sec. 7.5.2, pp. 52–53). Without drag the speed grows linearly; with quadratic drag it saturates at the terminal speed.*

INTERPRETATION AND LIMITING CASES

- **Long times:** $\tanh s \to 1$ as $s \to \infty$, so $v(t) \to w_T$: the body approaches terminal velocity but never exceeds it.
- **Short times:** $\tanh s = s - s^3/3 + \dots$, so $v(t) \approx w_T \cdot gt/w_T = gt$: the drag-free result of 5.2 is recovered while $v \ll w_T$.
- **Vanishing drag:** $k \to 0$ gives $w_T \to \infty$ and again $v \to gt$ for every fixed t (this is the limit Robinson asks for in Ex. 8.6).
- **Time scale:** w_T/g is the time to reach about 76% of w_T ($\tanh 1 \approx 0.76$); for the skydiver $54.2/9.8 \approx 5.5$ s.
- **Dimensions:** $[kv^2] = \text{kg m}^{-1} \cdot \text{m}^2\text{s}^{-2} = \text{N}$; $[\sqrt{mg/k}] = \sqrt{\text{kg m s}^{-2}/(\text{kg m}^{-1})} = \text{m/s}$; $[gt/w_T] = 1$.

AEROSPACE CONNECTION

In aerodynamic notation the drag is $D = \tfrac{1}{2}\rho\, C_D A\, v^2$, so $k = \tfrac{1}{2}\rho C_D A$ and $w_T = \sqrt{2mg/(\rho C_D A)}$. Terminal velocity grows as the air thins (ρ decreases with altitude), which is why a re-entering capsule or a high-altitude jumper falls much faster in the upper atmosphere. That refinement makes k depend on height and is beyond this topic.

7. Summary and lecture map

IDEA	KEY STATEMENT	CLASS NOTES	ROBINSON	LECTURE
ODE vs PDE	One independent variable vs several; $x' = x$, heat equation	p. 2	Sec. 3.1, p. 11	Sep 8
Order, autonomy	Highest derivative; no explicit t	p. 2 (autonomy p. 3)	Sec. 3.2, p. 13	Sep 8
Linearity	Form (L); superposition test: $u' = u$ linear, $u'' = (u')^2 + 1$ not	p. 2	Sec. 3.3, p. 13–14	Sep 8–10 (corrected)
Systems	Two or more dependent variables; $x' = 2x - y$, $y' = -x + 3y$	p. 2	Eq. (3.9), p. 15; Ch. 25	Sep 8
Population growth / decay	$x' = kx \Rightarrow x = x_0 e^{kt}$; grows if $k > 0$, decays if $k < 0$	p. 2	Sec. 8.2–8.3, p. 61–64	Sep 8
Logistic correction	$x' = kx(1 - x/M)$, $x \to M$; analysed in Topic 2	p. 2 (analysis p. 3)	Sec. 8.5, p. 66–68	Sep 8–10
Free fall	$-mg = my'' \Rightarrow y = -\frac{1}{2}gt^2$ (from rest at 0)	p. 2	Sec. 5.3, Ex. 5.6	Sep 8
Quadratic drag	$v = w_T \tanh(gt/w_T)$, $w_T = \sqrt{mg/k}$	–	Sec. 7.5.2, Ex. 8.6	–
IVP, Picard's theorem	Existence and uniqueness of the solution of an IVP	p. 3	Ch. 6	Sep 10 (Topic 2)

WHAT COMES NEXT

Topic 2 follows pages 3 and 4 of the class notes: initial value problems and Picard's theorem, autonomous equations and the stability of their stationary points, and three methods that give explicit solutions (separable equations, the integrating factor and exact equations).